

ERDŐS INSTITUTE: VIX PROJECT

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1. INTRODUCTION

The VIX is the Chicago Board Options Exchange’s measure of how much the market expects the S&P 500 to move, up or down, over the next 30 days. It is often called the “fear gauge” because it tends to spike during periods of market stress. From a mathematical perspective, it is interesting because it avoids committing to any particular story about how volatility behaves. Rather than fitting a model and asking the model what it thinks about volatility, the VIX is built directly out of prices of S&P 500 options themselves. It is a non-obvious fact that expected variance can be recovered from such prices. We give a self-contained derivation of the analytic formula for the VIX, starting from the dynamics of the underlying and ending at the discretized formula actually used to compute the published index from a real options chain.

This document is the theory companion to the accompanying Jupyter notebook, which implements the calculations described below: the VIX reconstruction, the Merton jump-diffusion calibration, and the resulting diagnostic comparisons. Our aim here is to derive and justify each formula the notebook uses and to describe what the notebook computes; the resulting figures, fitted parameter values, and empirical conclusions are reported in the notebook itself, not here.

1.1. Derivation of the analytic formula for VIX index. The price of the S&P 500 is modeled, under the risk-neutral measure, by the stochastic differential equation (SDE)

$$(1) \quad dS_t = rS_t dt + \sigma_t S_t dW_t,$$

where W_t is a standard Brownian motion, r is the (constant) risk-free rate, and σ_t is the instantaneous volatility, all at time t .

Assumption. We place no model on σ_t beyond requiring it to be a well-behaved, square-integrable random process: it may itself be stochastic, time-varying, and path-dependent in any way. The only structural restriction in Equation (1) is that S_t moves continuously, with no jumps; nothing about how volatility itself evolves is assumed. This is the precise sense in which the VIX is *model-free*: the formula we derive will depend only on observable option prices, not on which particular volatility model happens to be true of the market.

We want the risk-neutral expected total variance over $[0, T]$, $\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right)$, and the strategy is to relate this quantity to $\mathbb{E}(\log S_T)$, the expected log price at the fixed horizon T , since the latter turns out to be recoverable from option prices. Applying Itô’s lemma to $\log S_t$ in Equation (1),

$$(2) \quad d\log(S_t) = \frac{dS_t}{S_t} - \frac{(dS_t)^2}{2S_t^2} = r dt + \sigma_t dW_t - \frac{1}{2}\sigma_t^2 dt,$$

using $(dS_t)^2 = \sigma_t^2 S_t^2 dt$, which holds because S_t is driven only by Brownian motion with no jump component. Integrating both sides from $t = 0$ to $t = T$ gives

$$\log(S_T) - \log(S_0) = rT + \int_0^T \sigma_t dW_t - \frac{1}{2} \int_0^T \sigma_t^2 dt.$$

Solving for $\log(S_T)$ and taking the expectation,

$$\mathbb{E}(\log(S_T)) = \log(S_0) + rT + \mathbb{E}\left(\int_0^T \sigma_t dW_t\right) - \frac{1}{2}\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right).$$

Since $\int_0^T \sigma_t dW_t$ is a stochastic integral against Brownian motion, by the Itô isometry, it has expectation zero (note we need not assume σ_t be independent of W_t). Writing $F = S_0 e^{rT}$ for the risk-neutral forward price of the stock, $\log(S_0) + rT = \log F$, so

$$\mathbb{E}(\log(S_T)) = \log(F) - \frac{1}{2}\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right).$$

Rearranging, we get the following formula for the expectation of total realized variance over the interval $[0, T]$:

$$\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right) = 2(\log F - \mathbb{E}(\log S_T)).$$

Interpreting the quantity

$$\bar{\sigma}_T^2 := \frac{1}{T}\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right)$$

as the average annualized variance, we get

$$(3) \quad \bar{\sigma}_T^2 = \frac{2}{T}(\log F - \mathbb{E}(\log S_T)).$$

We have reduced the problem to computing a single number, $\mathbb{E}(\log S_T)$, the risk-neutral expected value of the log contract. The log contract itself is not a traded instrument, so the remaining task is to price it using instruments that are: puts and calls.

Now we use the following version of Taylor's theorem with integral remainder from calculus to rewrite $\log(S_T)$ in terms of the payoffs of put and call options.

Theorem 1.1. *Let $f(x)$ be a twice-differentiable function on (a, b) . Pick any x_0 in (a, b) . Then for any $x \in (a, b)$,*

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \int_a^{x_0} (y - x)^+ f''(y) dy + \int_{x_0}^b (x - y)^+ f''(y) dy,$$

where $z^+ = \max(0, z)$.

We apply Theorem 1.1 to $f(S) = \log S$ with $x_0 = K_0$, a fixed reference strike. Since $f'(S) = 1/S$ and $f''(S) = -1/S^2$, the two integral terms carry a minus sign:

$$\log S_T = \log(K_0) + \frac{S_T - K_0}{K_0} - \int_0^{K_0} \frac{(K - S_T)^+}{K^2} dK - \int_{K_0}^\infty \frac{(S_T - K)^+}{K^2} dK.$$

Taking risk-neutral expectations of both sides, and recognizing that the discounted expectation of a put or call payoff is exactly the time-0 price of that option, $e^{-rT}\mathbb{E}((K - S_T)^+) = P(K)$ and $e^{-rT}\mathbb{E}((S_T - K)^+) = C(K)$,

$$\mathbb{E}(\log S_T) = \log(K_0) + \frac{\mathbb{E}(S_T) - K_0}{K_0} - \int_0^{K_0} \frac{e^{rT}P(K)}{K^2} dK - \int_{K_0}^\infty \frac{e^{rT}C(K)}{K^2} dK,$$

where $P(K)$ is the time-0 price of the European-style put with strike K and $C(K)$ is the time-0 price of the European-style call with strike K . Using $\mathbb{E}(S_T) = F$, the risk-neutral forward price, and substituting into Equation (3), we obtain an analytic formula for the **average annualized variance** over $[0, T]$ in terms of

- the forward price of the underlying, F , and

- **integrals, weighted by $1/K^2$, of put and call prices across all strikes.**

$$(4) \quad \bar{\sigma}_T^2 = \frac{2}{T} \left(\log \left(\frac{F}{K_0} \right) - \left(\frac{F}{K_0} - 1 \right) + \int_0^{K_0} \frac{e^{rT} P(K)}{K^2} dK + \int_{K_0}^{\infty} \frac{e^{rT} C(K)}{K^2} dK \right).$$

The published VIX is $100\sqrt{\bar{\sigma}_T^2}$ (a volatility, expressed in percentage points; $\bar{\sigma}_T^2$ itself is a variance).

The $1/K^2$ weight on the option strip is forced on us by $f''(S) = -1/S^2$ in Theorem 1.1. It is worth pausing on why this works at all: we started by refusing to assume any model for σ_t , which made $\mathbb{E}(\log S_T)$ look unrecoverable from anything observable today, since it is defined in terms of the entire risk-neutral law of S_T . Theorem 1.1 is what rescues us, by writing $\log S_T$ exactly, for every possible realization of S_T and not merely in expectation, as a static portfolio: a forward position plus an integral of out-of-the-money puts below K_0 and calls above. Static positions held to expiration, with no further trading, are exactly what option prices observed today can tell us about.

1.2. The numerical approximation of the VIX formula. Equation (4) requires a continuum of strikes, one option for every $K \in (0, \infty)$, and is not directly computable from a real exchange, which lists options at only a finite set of strikes. Two approximations turn it into the formula actually used in practice.

First, since K_0 is chosen, in the next section, to be the listed strike nearest the forward F , the ratio F/K_0 is always close to 1. A second-order Taylor expansion of $\log(x)$ around $x = 1$ gives, for x near 1,

$$\log(x) - (x - 1) \approx -\frac{1}{2}(x - 1)^2,$$

so that

$$\log \left(\frac{F}{K_0} \right) - \left(\frac{F}{K_0} - 1 \right) \approx -\frac{1}{2} \left(\frac{F}{K_0} - 1 \right)^2.$$

This is a genuine approximation (it discards third- and higher-order terms in $F/K_0 - 1$), but a good one in practice, since K_0 is by construction chosen close to F .

Second, the continuum of strikes in the two integrals of Equation (4) is replaced by a finite sum over the actual listed strikes $K_1 < K_2 < \dots < K_n$, via a midpoint Riemann sum. Each strike K_i is assigned a width

$$\Delta K_i = \begin{cases} K_{i+1} - K_i, & i \text{ is the lowest listed strike,} \\ \frac{K_{i+1} - K_{i-1}}{2}, & K_i \text{ is an interior strike,} \\ K_i - K_{i-1}, & i \text{ is the highest listed strike,} \end{cases}$$

so that the strike axis is tiled with no gaps and no overlaps: each interior strike is given half the distance to each of its two neighbors, and the two edge strikes are given the one-sided gap to their single neighbor, since there is no further strike beyond the edge with which to split the interval. Writing $Q(K_i)$ for the price of the *out-of-the-money* option at K_i (a put if $K_i < K_0$, a call if $K_i > K_0$, and the average of the put and call quotes at $K_i = K_0$ itself), the two integrals are approximated by

$$\int_0^{K_0} \frac{e^{rT} P(K)}{K^2} dK + \int_{K_0}^{\infty} \frac{e^{rT} C(K)}{K^2} dK \approx \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i).$$

Remark. The out-of-the-money side is used at every strike, rather than calls or puts exclusively, because out-of-the-money options are the ones actually quoted competitively on a real

exchange. An in-the-money option's price is dominated by intrinsic value: the deterministic payoff from exercising immediately, which every market participant already agrees on. The genuinely volatility-sensitive part of its price is a small residual riding on top of that intrinsic value, so there is little for market makers to compete on and bid-ask spreads stay wide. An out-of-the-money option has zero intrinsic value, so its entire price reflects the market's view of future movement, which is exactly the information the replication argument needs and exactly what gets quoted tightly. Put-call parity guarantees this choice loses nothing relative to using the other side of the chain at every strike; it is a choice about which quote is reliable, not about the mathematics of Equation (4).

Combining both approximations, we obtain the discretized formula that is actually computed from a real options chain:

$$(5) \quad \bar{\sigma}_T^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2,$$

where $Q(K_i)$ is the mid-quote, the average of the highest bid and lowest ask of the relevant out-of-the-money option, and the forward price

$$F = K_0 + e^{RT}(C(K_0) - P(K_0))$$

is backed out from put-call parity at K_0 , the listed strike closest to the unobservable true forward. Equation (5) is exactly formula (1) of the Cboe Volatility Index Mathematics Methodology, and the VIX itself is $100\sqrt{\bar{\sigma}_T^2}$.

Remark. *This entire derivation rests on Equation (1) having no jump term: it is the no-jump assumption, not any assumption on σ_t , that lets Itô's lemma turn $(dS_t)^2$ into the clean term $\sigma_t^2 S_t^2 dt$ used above. If S_t is allowed to jump, this step picks up an extra contribution from the jump part of the quadratic variation, and Equation (3) no longer equals expected diffusive variance alone; it equals diffusive variance plus a jump term. The option-strip replication in Theorem 1.1 still prices the log contract correctly even with jumps, since that identity used no continuity assumption, but what the log contract's price represents in terms of variance changes once jumps are allowed. In other words, under jumps the VIX is still a well-defined, computable number, but it is a biased estimate of risk-neutral expected diffusive variance. This is the gap our jump-model extension is designed to address.*

2. THE MERTON MODEL: ISOLATING JUMP RISK

The remark above identifies exactly what the model-free construction cannot do: once S_t is allowed to jump, the log-contract replication still prices a well-defined number from the option strip, but that number is no longer purely $\mathbb{E} \left(\int_0^T \sigma_t^2 dt \right)$, the expected diffusive variance. There is an extra jump term, and Equation (4) and Equation (5) cannot, on their own, tell us how much of the published VIX is ordinary continuous volatility risk and how much is compensation for the fear of sudden crashes. Separating the two requires committing to a specific model of how jumps occur, since the model-free argument by construction throws away exactly the information needed to tell apart a diffusive move from a jump.

The simplest model that does this is due to Merton [1]: keep the price diffusion exactly as simple as it can be, a constant volatility, geometric Brownian motion, and add nothing but a compound Poisson jump term on top. We follow the treatment and notation of Cont and Tankov [2], Section 3.3 and Chapter 2, throughout this section.

2.1. Model specification. Following Cont and Tankov [2], we work directly with the log-price $X_t = \log S_t$, which under the risk-neutral measure follows the jump-diffusion

$$(6) \quad X_t = X_0 + \gamma t + \sigma W_t + \sum_{i=1}^{N_t} Y_i,$$

where W_t is a standard Brownian motion, $\sigma > 0$ is a constant volatility, N_t is a Poisson process of constant intensity λ independent of W_t , and $Y_1, Y_2, \dots$ are i.i.d. jump sizes, independent of both W_t and N_t , drawn from

$$Y_i \sim \mathcal{N}(m, \delta^2).$$

Each jump moves the log-price by Y_i , so the price itself jumps multiplicatively, $S_t = S_{t-} e^{Y_i}$ at a jump time.

In plain terms: σ plays exactly the role it does in Black-Scholes, controlling how much the price wiggles on an ordinary day with no jump involved. The parameter λ is the average number of jumps per year. The parameter m is the average size of a jump, measured in log-price units; we do not fix its sign ahead of time, since the data should tell us whether jumps tend to be drops, rises, or neither on average. The parameter δ measures how much individual jump sizes vary around that average. Finally, γ is just the ordinary drift of $\log S_t$ between jumps, playing the same role that $r - \sigma^2/2$ plays in Black-Scholes; we have not pinned it down yet, and doing so is the first order of business below.

This is the same compound Poisson specification used in Cont and Tankov's discussion of the Merton model as a Lévy process, with jump (Lévy) measure

$$\nu(dx) = \frac{\lambda}{\delta\sqrt{2\pi}} e^{-(x-m)^2/2\delta^2} dx,$$

a finite measure of total mass λ , reflecting that jumps arrive at the constant rate λ and, conditional on a jump occurring, the jump size is Gaussian. Four parameters, $(\sigma, \lambda, m, \delta)$, together with the drift γ , fully specify the model; γ itself is fixed by the requirement that the discounted price be a risk-neutral martingale, which we derive next.

2.2. The martingale (risk-neutral) drift condition. For $S_t = e^{X_t}$ to be, after discounting, a risk-neutral martingale, we need $\mathbb{E}(e^{-rt} S_t) = S_0$ for all t , i.e. $\mathbb{E}(e^{X_t}) = S_0 e^{rt}$. Since W_t , N_t , and the Y_i are mutually independent, we can compute $\mathbb{E}(e^{X_t})$ by conditioning on the number of jumps $N_t = n$ and using independence:

$$\mathbb{E}(e^{X_t}) = e^{X_0 + \gamma t} \mathbb{E}(e^{\sigma W_t}) \mathbb{E}\left(e^{\sum_{i=1}^{N_t} Y_i}\right).$$

The first expectation is the standard log-normal moment generating function, $\mathbb{E}(e^{\sigma W_t}) = e^{\sigma^2 t/2}$. For the second, conditioning on $N_t = n$ and using that the Y_i are i.i.d. and independent of N_t ,

$$\mathbb{E}\left(e^{\sum_{i=1}^{N_t} Y_i}\right) = \sum_{n=0}^{\infty} \mathbb{P}(N_t = n) (\mathbb{E}(e^{Y_1}))^n = \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \left(e^{m + \delta^2/2}\right)^n = \exp\left(\lambda t \left(e^{m + \delta^2/2} - 1\right)\right),$$

where we used $\mathbb{E}(e^{Y_1}) = e^{m + \delta^2/2}$, the moment generating function of a $\mathcal{N}(m, \delta^2)$ random variable evaluated at 1, and recognized the resulting sum as the moment generating function of the Poisson distribution. Writing

$$(7) \quad k := e^{m + \delta^2/2} - 1$$

for the expected proportional jump size, the martingale condition $\mathbb{E}(e^{X_t}) = S_0 e^{rt}$ becomes

$$X_0 + \gamma t + \frac{\sigma^2 t}{2} + \lambda t k = X_0 + rt,$$

so that

$$(8) \quad \gamma = r - \frac{\sigma^2}{2} - \lambda k.$$

With this choice of γ , Equation (6) is exactly Cont and Tankov's risk-neutral Merton model, and $e^{-rt}S_t$ is a martingale by construction.

2.3. The variance decomposition. We now derive the central formula of this section: the split of total expected variance into a piece coming from the continuous part of Equation (6) and a piece coming from the jumps.

First, a bridge to Section 1's notation. There, $\bar{\sigma}_T^2$ was defined, just before Equation (3), as $\frac{1}{T}\mathbb{E}\left(\int_0^T \sigma_t^2 dt\right)$, and Itô's lemma turned $(dS_t)^2$ into $\sigma_t^2 S_t^2 dt$ precisely because S_t , and hence $\log S_t$, moved continuously. In the language of stochastic calculus, $\int_0^T \sigma_t^2 dt$ is the *quadratic variation* of $\log S_t$ over $[0, T]$, which we write $\langle \log S \rangle_T$. Concretely: take a partition $0 = t_0 < t_1 < \dots < t_n = T$ of $[0, T]$, form the sum of squared increments $\sum_{j=0}^{n-1} (X_{t_{j+1}} - X_{t_j})^2$, and let the mesh of the partition, the width of its widest subinterval, shrink to zero. $\langle \log S \rangle_T$ is defined to be the limit of these sums. We use the same name and the same limiting definition for $\log S_t$ under the Merton model Equation (6); the only difference is that $\log S_t$ is no longer continuous, so the sum of squared increments no longer converges to $\int_0^T \sigma_t^2 dt$ alone.

To see what it converges to instead, partition $[0, T]$ into n subintervals of width T/n and consider $\sum_j (X_{t_{j+1}} - X_{t_j})^2$. Since N_T is almost surely finite (a Poisson process with finite intensity has finitely many arrivals on any bounded interval), once the partition is fine enough every jump falls in its own subinterval, with no two jumps sharing one. A subinterval containing one jump of size Y_i has increment $Y_i + O(\sqrt{T/n})$, the jump itself plus a vanishingly small continuous piece, so its squared increment converges to Y_i^2 as $n \rightarrow \infty$. Every other subinterval contains no jump, so its increment is purely $\gamma(T/n) + \sigma(W_{t_{j+1}} - W_{t_j})$, and the sum of these squared increments converges, by the usual argument for the quadratic variation of Brownian motion, to $\sigma^2 T$, unaffected by removing the finitely many, vanishing-width jump subintervals from the sum. Adding the two limits gives the quadratic variation of $\log S_t$ under Equation (6),

$$\langle \log S \rangle_T = \sigma^2 T + \sum_{i=1}^{N_T} Y_i^2,$$

with no cross term between the two pieces, since in the limit of a fine partition every subinterval is asymptotically either a jump interval or a continuous interval, never both. (This is a special case of the calculation of the quadratic variation of any semimartingale; see Cont and Tankov [2], Chapter 8, for that fully general version.)

Normalizing to a per-unit-time quantity, exactly as $\bar{\sigma}_T^2$ normalizes $\int_0^T \sigma_t^2 dt$ above, total realized variance under the Merton model over $[0, T]$ is

$$(9) \quad RV(0, T) = \frac{1}{T} \langle \log S \rangle_T = \frac{1}{T} \left(\sigma^2 T + \sum_{i=1}^{N_T} Y_i^2 \right) = \underbrace{\sigma^2}_{\text{continuous}} + \underbrace{\frac{1}{T} \sum_{i=1}^{N_T} Y_i^2}_{\text{jump}}.$$

The continuous piece is already a constant, σ^2 , with nothing left to take an expectation of. The jump piece requires one further step: N_T itself is random, so $\sum_{i=1}^{N_T} Y_i^2$ is a random sum of a random number of terms. We compute its expectation by conditioning on N_T , exactly as in the

drift-condition calculation above. Since the Y_i are i.i.d. and independent of N_T ,

$$\mathbb{E}\left(\sum_{i=1}^{N_T} Y_i^2 \middle| N_T = n\right) = n \mathbb{E}(Y_1^2),$$

and by the tower property, taking expectation of both sides over N_T , whose mean is $\mathbb{E}(N_T) = \lambda T$ since N_T is Poisson with parameter λT ,

$$\mathbb{E}\left(\sum_{i=1}^{N_T} Y_i^2\right) = \mathbb{E}\left(\mathbb{E}\left(\sum_{i=1}^{N_T} Y_i^2 \middle| N_T\right)\right) = \mathbb{E}(N_T) \mathbb{E}(Y_1^2) = \lambda T \cdot \mathbb{E}(Y_1^2).$$

Since $Y_1 \sim \mathcal{N}(m, \delta^2)$, its second moment is $\mathbb{E}(Y_1^2) = \text{Var}(Y_1) + (\mathbb{E}(Y_1))^2 = \delta^2 + m^2$. Substituting,

$$\mathbb{E}\left(\frac{1}{T} \sum_{i=1}^{N_T} Y_i^2\right) = \lambda (m^2 + \delta^2).$$

Taking expectations of Equation (9) term by term and combining the two pieces gives the Merton analogue of Equation (3), the central quantity of this section:

$$(10) \quad \bar{\sigma}_{T, \text{Merton}}^2 := \mathbb{E}(RV(0, T)) = \underbrace{\sigma^2}_{\text{diffusive component}} + \underbrace{\lambda (m^2 + \delta^2)}_{\text{jump component}}.$$

Every quantity on the right-hand side is one of the four calibrated model parameters or a simple function of them. There is no remaining integral or numerical approximation, in contrast to the discretized VIX formula Equation (5), precisely because we have committed to a specific parametric model of how jumps occur. This is the price of the decomposition: the model-free VIX needed no assumption beyond continuity to produce Equation (5), while Equation (10) is only as good as the Merton specification itself.

2.4. Calibrating the model to the data we have. Everything above is a statement about the model in the abstract. To actually put a number on $\bar{\sigma}_{T, \text{Merton}}^2$ for a given day, we need values of $(\sigma, \lambda, m, \delta)$ that are consistent with that day's option prices. To carry out this calibration, we use the data already assembled for the model-free VIX calculation in Section 1: for a fixed quote date and a fixed expiration T , we have the forward price F , the reference strike K_0 , and the out-of-the-money mid-quotes $Q(K_i)$ at every listed strike K_i , which are the same inputs that feed into Equation (5).

To compare model prices against $Q(K_i)$ on a common footing, we first convert every quote to its call-equivalent price $C^*(K_i)$: quotes above K_0 already are call prices, and quotes below K_0 are converted via put-call parity, $C^*(K_i) = Q(K_i) + e^{-rT}(F - K_i)$. Calibration then asks for the parameter vector $\theta = (\sigma, \lambda, m, \delta)$ that minimizes a weighted sum of squared pricing errors across the chain,

$$(11) \quad \hat{\theta} = \arg \min_{\theta} \sum_i w_i (C_{\theta}(K_i, T) - C^*(K_i))^2, \quad w_i \propto \frac{\Delta K_i}{K_i^2},$$

where $C_{\theta}(K, T)$ is the price the Merton model itself assigns to a call struck at K , given parameters θ , and the weights w_i are normalized to have mean one across the strikes used on a given day. These are the same weights used to build the discretized VIX integral in Equation (5). Without them, a raw dollar-price objective is dominated by near-the-money strikes whose dollar prices are largest, and gives little weight to the far out-of-the-money strikes that carry most of the information about tail and jump risk.

We need $C_\theta(K, T)$ in closed form for Equation (11) to be something a computer can actually minimize. We use the same trick of conditioning on the number of jumps $N_T = n$. Here it turns the call price into a Poisson-weighted mixture of ordinary Black-Scholes prices,

$$(12) \quad C_\theta(K, T) = \sum_{n=0}^{\infty} e^{-\lambda T} \frac{(\lambda T)^n}{n!} C_{BS}\left(S_0(1+k)^n e^{-\lambda k T}, K, T, r, \sigma_n\right), \quad \sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T},$$

with $k = e^{m+\delta^2/2} - 1$ the same expected proportional jump size from Equation (7). Conditional on exactly n jumps having occurred, the price behaves like an ordinary Black-Scholes price, but at an adjusted effective spot, $S_0(1+k)^n e^{-\lambda k T}$, reflecting the average extra move the n jumps contributed, and an adjusted volatility σ_n that absorbs the extra variance those n jumps added over the period. Weighting each of these Black-Scholes prices by the Poisson probability of seeing exactly n jumps and summing over n gives the unconditional price. In practice the sum converges quickly and is truncated once its terms become negligible, since λT is small for a single expiration.

2.5. An identification problem, and how we resolve it. Equation (11) is, in principle, a nonlinear least-squares problem in four unknowns, solved numerically subject to the natural bounds $\sigma > 0$, $\lambda \geq 0$, $\delta > 0$. In practice, fitting all four parameters to a single day's chain does not work well: λ , m , and δ all affect the tails of the implied distribution in overlapping ways, so a rare, large jump and a frequent, modest jump can produce very similar option prices on any one day. A single day's chain therefore underdetermines the four parameters jointly; the optimizer finds many different (λ, m, δ) combinations that fit almost equally well, rather than one clear answer.

We resolve this by separating the parameters that are stable across market conditions from the ones that are not. The typical size and dispersion of a jump, m and δ , are treated as structural constants and estimated once from a pooled sample of dates rather than re-estimated every day; the jump frequency λ and the diffusive volatility σ , which we do expect to move with market conditions, are left free to vary daily. Concretely, we first select a set of representative dates $\mathcal{D}$, chosen to span the full observed range of VIX levels, and solve the nested problem

$$(13) \quad (\hat{m}, \hat{\delta}) = \arg \min_{m, \delta} \sum_{d \in \mathcal{D}} \min_{\sigma_d, \lambda_d} \sum_i w_i^{(d)} \left(C_{(\sigma_d, \lambda_d, m, \delta)}(K_i^{(d)}, T^{(d)}) - C^*(K_i^{(d)}) \right)^2,$$

an outer search over (m, δ) in which, for each candidate value, an inner fit of (σ_d, λ_d) is solved separately for every pooled date d , and the total pricing error across all of $\mathcal{D}$ is what the outer search minimizes. Because this pools option prices across many days to pin down only two unknowns, rather than one day's prices trying to pin down four, it is a substantially better-posed problem than fitting all four parameters at once.

With $(\hat{m}, \hat{\delta})$ fixed at the resulting pooled estimate, we return to Equation (11) and solve it once more for every quote date and expiration in the full sample, now only over (σ, λ) , using the previous day's result as an initial guess for each new day's fit for stability. This gives the daily time series $\hat{\sigma}_t$ and $\hat{\lambda}_t$, with $(\hat{m}, \hat{\delta})$ held in common across the whole 2018–2023 sample. Substituting each day's $(\hat{\sigma}_t, \hat{\lambda}_t, \hat{m}, \hat{\delta})$ into Equation (10) gives that day's diffusive component, $\hat{\sigma}_t^2$, and jump component, $\hat{\lambda}_t(\hat{m}^2 + \hat{\delta}^2)$, which is what the notebook plots and compares against the rest of the data described next.

2.6. Diagnostics and comparisons computed in the notebook. With a daily $(\hat{\sigma}_t, \hat{\lambda}_t, \hat{m}, \hat{\delta})$ series in hand, the notebook runs four comparisons.

First, before drawing any conclusion from the split itself, the notebook checks the fitted total implied variance, $\hat{\sigma}_t^2 + \hat{\lambda}_t(\hat{m}^2 + \hat{\delta}^2)$ from Equation (10), against the model-free VIX computed

from that same chain in Section 1. Both are estimates of the same quantity, total risk-neutral implied variance, under different assumptions; the notebook regresses one against the other across the full sample and reports the resulting slope, intercept, and R^2 . This is a check on the calibration itself, not on the economic hypothesis below: persistent disagreement between the two would be a sign that a day’s fit, or the pooled $(\hat{m}, \hat{\delta})$ estimate, needs attention before the diffusive/jump split built on top of it is trusted.

Second, the notebook plots four series together over the full sample: the raw VIX, forward realized volatility (a 21-trading-day rolling standard deviation of SPX log-returns, shifted forward one month so that each date’s value reflects what was actually realized over the following month), and the two Merton components, $\hat{\sigma}_t$ and $\sqrt{\hat{\lambda}_t(\hat{m}^2 + \hat{\delta}^2)}$. Plotting all four together lets the diffusive and jump components be compared directly against both the market-implied and the realized measures of volatility.

Third, several well-known stress episodes within the sample are marked directly on this time series: the February 2018 volatility-product unwind (“Volmageddon”), the fourth-quarter 2018 sell-off, the COVID-19 crash of early 2020, and the 2022 rate-hike bear market. Marking these dated windows lets the behavior of the jump and diffusive components around a specific, named event be read directly off the plot.

Fourth, the notebook computes the jump share of total variance, $\hat{\lambda}_t(\hat{m}^2 + \hat{\delta}^2)/(\hat{\sigma}_t^2 + \hat{\lambda}_t(\hat{m}^2 + \hat{\delta}^2))$, for every date, and summarizes its mean and median within calm, elevated, and stressed VIX regimes (thresholds of 20 and 30, respectively).

These comparisons are descriptive. A formal predictive test of whether the diffusive component forecasts realized volatility more accurately than the raw VIX, the sharpest version of the project’s underlying hypothesis, is a natural next step but is not yet implemented. All fitted parameters, figures, and conclusions drawn from the comparisons above are reported in the notebook itself; this summary, and the longer companion write-up it accompanies, are limited to motivating and deriving the methodology.

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